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Detection of hidden antibiotic resistance through real-time genomics

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Detection of hidden antibiotic resistance through real-time genomics

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Abstract

Real-time genomics through nanopore sequencing holds the promise of fast antibiotic resistance prediction, but concerns remain about its accuracy. We here show that real-time genomics has the potential to improve clinical practice based on a multi-drug resistant *Klebsiella pneumoniae* bloodstream infection. In contrast to established diagnostics, real-time genomics can accurately identify complex antibiotic resistance patterns conferred through newly discovered low-abundance resistance genes, potentially significantly impacting clinical decision-making and patient outcome.

Main

The World Health Organization has declared antibiotic resistance one of the ten most severe global health threats[1], with resistant infections leading to higher mortality and morbidity due to delayed or inappropriate therapy[2]. The rapid and accurate identification of resistant bacterial pathogens could improve patient outcomes by facilitating the earlier administration of appropriate therapy. Real-time genomics through nanopore sequencing technology has the potential of speeding up pathogen and antibiotic resistance profiling in the clinical setting[3] (**Figure 1**). The combination of real-time analysis and the technology's portability for *in situ* data creation allows for cost-efficient adaptive application, where as much genomic data as necessary can be obtained to reach minimum certainty thresholds in timely, clinically relevant predictions[4]. Long nanopore sequencing reads can further be leveraged to

create highly accurate, near-complete genome assemblies for strain-level identification and *de novo* detection of bacterial pathogens and their antibiotic resistance profiles[5]. This is especially relevant for complex infections where clinically established methods for taxonomic bacterial identification (*e.g.*, MALDI-TOF MS) and resistance profiling (*e.g.*, VITEK2; **Methods**; **Figure 1**) can lack resolution and where rapid targeted therapy can be particularly beneficial. However, concerns remain about the accuracy of real-time genomics for phenotypic antibiotic resistance prediction in comparison to established diagnostic approaches. We here showcase the potential power of *in situ* real-time genomics in the clinical setting through the example of a *Klebsiella pneumoniae* infection that harbors a newly discovered antibiotic resistance gene in low abundance.

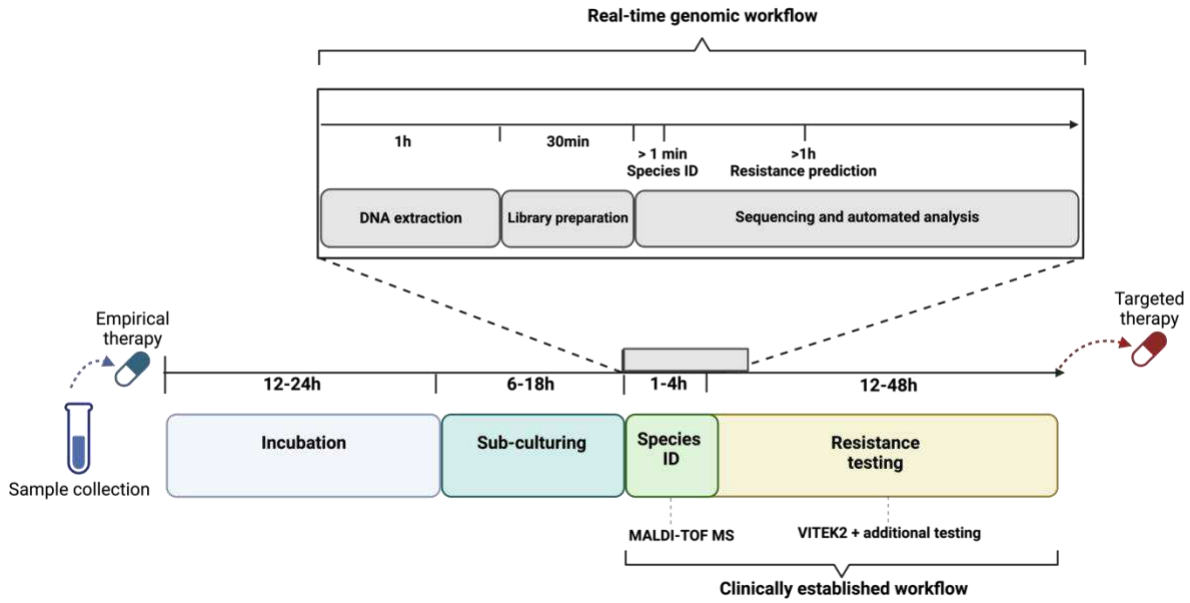


Figure 1. Workflow overview of real-time genomic (*top*) and clinically established (*bottom*) diagnostic approaches for pathogen species identification and antibiotic resistance profiling. After incubation and primary pathogen identification, pure bacterial isolates are recovered through sub-culturing, followed by pathogen and resistance profiling of the isolates. While the clinically established workflow can take up to 52 hours after subculturing, the real-time genomic workflow delivers first data after ~1.5 hours and can be employed in an adaptive manner to create the necessary amount of data.

Briefly, an immunocompromised patient presented with fever at the University Hospital rechts der Isar in Germany, and was initially treated with the carbapenem Meropenem (**Figure 2**). Clinically established diagnostic approaches detected a carbapenem-resistant *K. pneumoniae* isolate in an endotracheal aspirate with KPC (*K. pneumoniae* carbapenemase) as the putative resistance-conferring mechanism (**Figure 2**; **Methods**). As such KPC-producing bacteria have become a significant resistance burden due to their ability to hydrolyze a variety of beta-lactam antibiotics including carbapenems[6], Ceftazidime-Avibactam (CAZ-AVI) has been introduced as a new treatment option[7]. Accordingly, CAZ-AVI was administered to our patient and susceptibility was confirmed by additional phenotypic testing (**Figure 2**; **Methods**). After initial clinical improvement, the patient deteriorated under CAZ-AVI

therapy, and subsequent blood cultures grew a *K. pneumoniae* isolate which was now susceptible to carbapenems but resistant to CAZ-AVI. While new KPC-variants conferring resistance to CAZ-AVI are known to have emerged[8–10], no carbapenemase could be identified in our post-treatment isolate using clinically established approaches and Meropenem was additionally administered. However, the patient passed away shortly after (**Figure 2**).

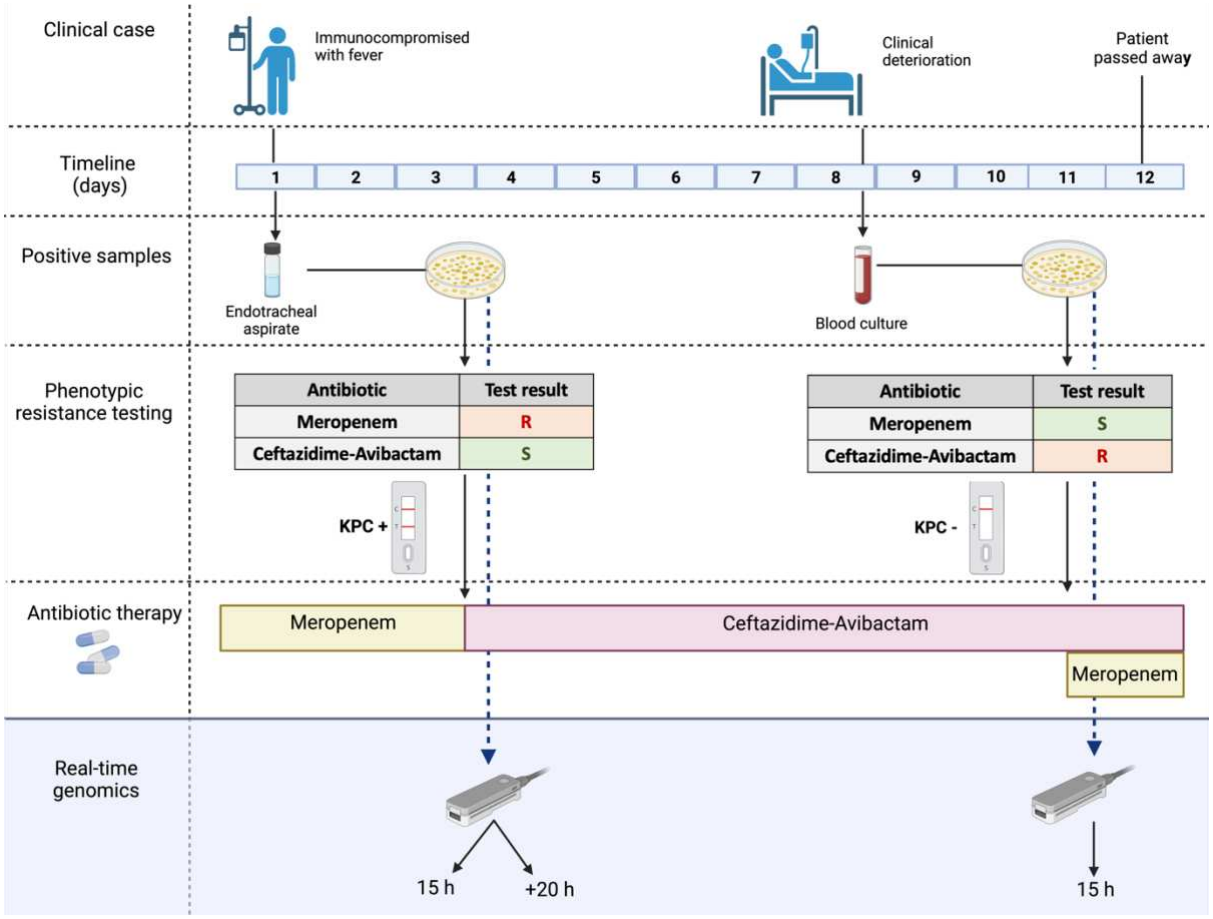


Figure 2. Overview of the course and treatment of the infection case. The patient was firstly treated with Meropenem. Bacterial isolates of the first positive patient sample (endotracheal aspirate) were subjected to VITEK2 for general resistance testing and additional tests for CAZ-AVI resistance and KPC detection (R: resistant; S: susceptible; KPC+/-: absence or presence of KPC; Methods); the respective results led to a change in the antibiotic treatment to CAZ-AVI after three days. After clinical deterioration, the second positive patient sample (blood culture) showed reversed antibiotic resistance test results. While Meropenem was subsequently administered, the patient passed away shortly after. We used real-time genomics to sequence DNA from the pre- and post-CAZ-AVI treatment isolates using the portable nanopore sequencing device Mk1b. Both isolates were sequenced for 15 hours, and the first isolate was sequenced for another 20 hours to simulate the potential adaptive sequencing in the clinical setting.

We simulated the retrospective clinical application of real-time genomics for antibiotic resistance prediction for this infection case as follows: We applied nanopore shotgun sequencing (Oxford Nanopore Technologies) to both *K. pneumoniae* isolates (pre- and post-CAZ-AVI treatment) by using the portable Mk1b sequencing device, rapid barcoding library preparation, and R9 nanopore

chemistry (**Figure 2**; **Methods**). We processed the raw nanopore data using high-accuracy Guppy basecalling and flye, minimap2, and Racon for *de novo* genome assembly and polishing. We used EPI2ME WIMP for species classification, and EPI2ME Antimicrobial Resistance's protein homolog model for antibiotic resistance predictions[11] (**Supplementary Figure 1**; **Methods**). We additionally simulated the adaptive application of real-time genomics in the clinic, where the same patient sample can be sequenced as long as necessary to obtain more data and certainty on resistance predictions. We here sequenced the pre-treatment isolate for additional 20 hours (**Figure 2**; **Methods**).

Using this real-time genomic data, we correctly identified *K. pneumoniae* as the dominant pathogen in the pre- and post-treatment isolates. In the pre-treatment isolate, we detected the presence of accurate (> 90%) KPC-2 gene copies (n=40; **Table 1**), confirming the phenotypic Meropenem resistance (**Figure 2**)[12]. In the post-treatment isolate, we detected many copies of the KPC-14 gene (n=44; **Table 1**), which had previously been described as one of the few KPC subtypes that confer CAZ-AVI resistance and carbapenem susceptibility[9]. The genomics-based resistance prediction therefore accurately predicted the phenotypic antibiotic resistance, and it additionally identified KPC-14 as the putative resistance-conferring mechanism (**Figure 1**).

Our real-time genomics approach importantly also detected one copy of the KPC-14 resistance gene in the pre-treatment isolate; while this one copy would have been too little evidence to predict phenotypic CAZ-AVI resistance, our simulation of the adaptive application of real-time genomics in the clinic showed us that less than 20 hours of additional sequencing would have resulted in the detection of four more highly accurate copies of KPC-14 (**Table 1**). The second KPC-14 gene copy would have already been detected after two hoursof additional sequencing and could have rapidly influenced antibiotic treatment through the early detection of CAZ-AVI resistance potential (**Supplementary Figure 2**; **Methods**).

Table 1. Real-time genomic antibiotic resistance predictions using EPI2ME computational analysis from patient samples before and after CAZ-AVI treatment (R: resistant; S: susceptible; CN: copy-number of respective resistance genes; **Methods**).

Sequencing run	Meropenem				Ceftazidime-Avibactam			
	Prediction	Evidence	Accuracy	CN	Prediction	Evidence	Accuracy	CN
Pre-treatment 15h	R	KPC-2	93.7%	40	R?	KPC-14	92.6%	1
+20h	R		94.4%	147	R		96.3%	4
Post-treatment 15h	S	/	/	/	R	KPC-14	93.4%	44

We further analyzed our *de novo* assemblies using the Pathogenwatch[12] platform which enables clustering of core-genome multilocus sequence types, and serves as a reference for complex *K.*

pneumoniae lineages (**Methods**). We found that the pre- and post-treatment isolates consisted of the same emerging high-risk sequence type ST147[15]. Functional annotation of the assembled plasmid genomes using BugSeq[16] revealed that KPC-2 and KPC-14 were both located on IncN plasmids, which showed 99.7% identity according to sequence alignments (**Supplementary Figure 3**; **Methods**). These results suggest that the observed change in antibiotic resistance might have been caused by a complex infection of the same *K. pneumoniae* lineage with a low-abundance KPC-14-carrying IncN plasmid which became dominant due to its evolutionary selective advantage under CAZ-AVI exposure. Our genomic analyses further showed that we had actually discovered a new KPC resistance gene, KPC-159, that is very similar to the KPC-14 gene detected by our analysis (amino acid sequence similarity of 99.9%) and therefore conferred the same phenotypic antibiotic resistance (**Methods**).

The application of real-time genomics to this patient's case indicates the profound potential of real-time genomics for rapidly and accurately profiling complex bacterial infections in the clinical setting. Our findings show that nanopore sequencing could have detected the phenotypically hidden CAZ-AVI resistance, which would have directly impacted antibiotic therapy, such as through the early administration of alternative antibiotics or combination therapy[17]. Further, the genomics-based identification of a KPC gene as the underlying CAZ-AVI resistance mechanism would have directly informed clinical infection prevention protocols to reduce the risk of between- and within-patient KPC transmission. The fast, adaptive, and *in situ* nature of antibiotic resistance profiling by nanopore sequencing would have therefore surpassed current clinical practice in accurately informing clinical management. Importantly, we could leverage our genomic data to trace the change in antibiotic resistance back to low copy-number plasmid variation in the patient's original infection. The presence of multiple plasmids in an infection is a general public health hazard since it can accelerate the emergence of resistance when exposed to selection pressures[18,19]. This has previously been described as an underlying mechanism of the fast emergence of CAZ-AVI-resistance under CAZ-AVI exposure[19]. Given the expected high rate of antibiotic resistance emergence in the future, the demonstrated insufficiency of clinically established resistance detection approaches to accurately assess complex infections could pose a significant threat to efficient antibiotic treatment.

While genomics-based predictions might not be able to replace phenotypic resistance assertions, we anticipate great potential of combining the advantages of phenotypic with real-time genomic technology for antibiotic resistance profiling in the clinical setting. We further envision that the low upfront investment requirements of nanopore sequencing could provide guidance on antibiotic

therapy globally, especially in low- and middle-income settings where expensive machinery for standard clinical diagnostics are likely to not be accessible in the foreseeable future.

References

1. EClinicalMedicine. Antimicrobial resistance: a top ten global public health threat. *EClinicalMedicine* **41**, 101221 (2021).
2. Murray, C. J. L. *et al.* Global burden of bacterial antimicrobial resistance in 2019: a systematic analysis. *The Lancet* **399**, 629–655 (2022).
3. Taxt, A. M., Avershina, E., Frye, S. A., Naseer, U. & Ahmad, R. Rapid identification of pathogens, antibiotic resistance genes and plasmids in blood cultures by nanopore sequencing. *Sci Rep* **10**, 7622 (2020).
4. Bloomfield, M. *et al.* Oxford nanopore next generation sequencing in a front-line clinical microbiology laboratory without on-site bioinformaticians. *Pathology* (2023)
5. Foster-Nyarko, E. *et al.* Nanopore-only assemblies for genomic surveillance of the global priority drug-resistant pathogen, *Klebsiella pneumoniae*. *Microb Genom* **9**, (2023).
6. Hammoudi Halat, D. & Ayoub Moubareck, C. The Current burden of Carbapenemases: Review of significant properties and dissemination among Gram-negative bacteria. *Antibiotics* **9**, 186 (2020).
7. Lahiri, S. D. *et al.* Avibactam and Class C β-Lactamases: Mechanism of inhibition, conservation of the binding pocket, and implications for resistance. *Antimicrob Agents Chemother* **58**, 5704–5713 (2014).
8. Wu, Y. *et al.* Identification of a KPC variant conferring resistance to Ceftazidime-Avibactam from ST11 Carbapenem-resistant *Klebsiella pneumoniae* strains. *Microbiol Spectr* **10**, (2022).
9. Niu, S. *et al.* A Ceftazidime-Avibactam-resistant and Carbapenem-susceptible *Klebsiella pneumoniae* strain harboring *bla* $_{KPC-14}$ Isolated in New York City. *mSphere* **5**, (2020).
10. Shi, Q. *et al.* Multiple Novel Ceftazidime-Avibactam-resistant variants of *bla* $_{KPC-2}$ -positive *Klebsiella pneumoniae* in two patients. *Microbiol Spectr* **10**, (2022).
11. Alcock, B. P. *et al.* CARD 2023: expanded curation, support for machine learning, and resistome prediction at the Comprehensive Antibiotic Resistance Database. *Nucleic Acids Res* **51**, D690–D699 (2023).
12. Mehta, S. C. *et al.* KPC-2 β-lactamase enables carbapenem antibiotic resistance through fast deacylation of the covalent intermediate. *Journal of Biological Chemistry* **296**, 100155 (2021).
13. Argimón, S. *et al.* Rapid genomic characterization and global surveillance of *Klebsiella* using Pathogenwatch. *Clinical Infectious Diseases* **73**, S325–S335 (2021).
14. Lam, M. M. C. *et al.* A genomic surveillance framework and genotyping tool for Klebsiella pneumoniae and its related species complex. *Nat Commun* **12**, 4188 (2021).
15. Peirano, G., Chen, L., Kreiswirth, B. N. & Pitout, J. D. D. Emerging antimicrobial-resistant high-risk *Klebsiella pneumoniae* clones ST307 and ST147. *Antimicrob Agents Chemother* **64**, (2020).
16. Chandrakumar, I. *et al.* BugSplit enables genome-resolved metagenomics through highly accurate taxonomic binning of metagenomic assemblies. *Commun Biol* **5**, 151 (2022).
17. Lombardo, D., Ambretti, S., Lazzarotto, T. & Gaibani, P. In vitro activity of imipenem-relebactam against KPC-producing *Klebsiella pneumoniae* resistant to ceftazidime-avibactam and/or meropenem-vaborbactam. *Clinical Microbiology and Infection* **28**, 749–751 (2022).
18. Diaz Caballero, J. *et al.* Mixed strain pathogen populations accelerate the evolution of antibiotic resistance in patients. *Nat Commun* **14**, 4083 (2023).
19. Bianco, G., Boattini, M., Iannaccone, M., Cavallo, R. & Costa, C. Bloodstream infection by two subpopulations of *Klebsiella pneumoniae* ST1685 carrying KPC-33 or KPC-14 following ceftazidime/avibactam treatment: considerations regarding acquired heteroresistance and choice of carbapenemase detection assay. *Journal of Antimicrobial Chemotherapy* **75**, 3075–3076 (2020).

Methods

Clinically established species identification and antibiotic susceptibility testing

The clinically established microbiological workflow consisted of plating out clinical samples on BD® Columbia Blood and MacConkey agar plates (Beck Dickinson GmbH, Heidelberg, Germany), which were then incubated for approximately 16-24 hours. For blood cultures or mixed cultures, as was the case in the endotracheal aspirate, a subculturing step was added, after which species identification was done from one colony forming unit (CFU) of pure bacterial cultures using Matrix-assisted laser desorption ionization time-of-flight mass spectrometry instructions (MALDI-TOF MS, Bruker Daltronics GmbH, Leipzig Germany) according to the manufacturer's instructions. Antibiotic susceptibility testing was performed using VITEK2 (BioMérieux, Marcy l'Etoile, France). For this, up to three bacterial CFUs were transferred to a saline tube to generate a homogenous suspension with a density equivalent to 0.5 McFarland. Subsequently, Minimum Inhibitory Concentrations (MICs in mg/L) of the two isolates were measured and interpreted with the VITEK2 gram-negative (AST-GN69) card according EUCAST (European Committee on Antimicrobial Susceptibility Testing) guidelines[20] (**Supplementary Table 1**). Of note, the AST-GN69 card does not detect the presence of carbapenemases or measure MICs for Ceftazidime-Avibactam (CAZ-AVI). Thus, for isolates with antibiotic susceptibility profiles indicative of carbapenem resistance, we identified carbapenemases with a multiplex immunochromatography assay consisting of lateral flow assays (O.K.N.V.I Resist-5[21]). This test detects the presence of Oxa-48, NDM, VIM, IMP-carbapenemases and the most prevalent KPC subtypes (namely, the subtypes KPC-2, KPC-3, KPC-4) in a bacterial isolate[21]. In addition, the CAZ-AVI MIC was measured with the Liofilchem® MIC test strip (MTS™) (LIOFILCHEM s.r.l., Roseto degli Abruzzi, Italy). The test strip contains Ceftazidime concentrations ranging from 0,016-256 ug/ml with a fixed Avibactam concentration of 4ug/ml. Similar to the VITEK2 MIC measurements, up to three bacterial CFUs were transferred to a sterile saline tube to form a homogenised suspension with a density equivalent to 0.5 McFarland. This solution was then plated out on a BD® Muller Hinton Agar using sterile cotton-tipped swabs and incubated with a Ceftazidime-Avibactam Liofilchem® MTS™ for 16 hours. Again, MICs were interpreted following EUCAST guidelines (**Supplementary Table 1**). After completion of the phenotypic processing, the isolates were stored at -80°C.

Supplementary Table 1. MIC measurements with VITEK2 and Liofilchem®MIC test strip for CAZ-AVI. The colouring scheme indicates the interpretation of MICs according to EUCAST guidelines (green: susceptible/intermediate; red: resistant).

Antibiotic	Pre-treatment isolate MIC (mg/L)	Post-treatment isolate MIC (mg/L)
Ampicillin	>32	>32
Ampicillin-Sulbactam	>32	>32
Piperacillin	>128	>128
Piperacillin-Tazobactam	>128	>128
Cefuroxime	>64	>64
Cefotaxime	>64	>64
Ceftriaxone	>64	>64
Ceftazidime	>64	>64
Imipenem	>16	1
Meropenem	>16	2
Ertapenem	>8	>8
Gentamicin	>16	>16
Ciprofloxacin	>4	>4
Ceftazidime-Avibactam	2	>256

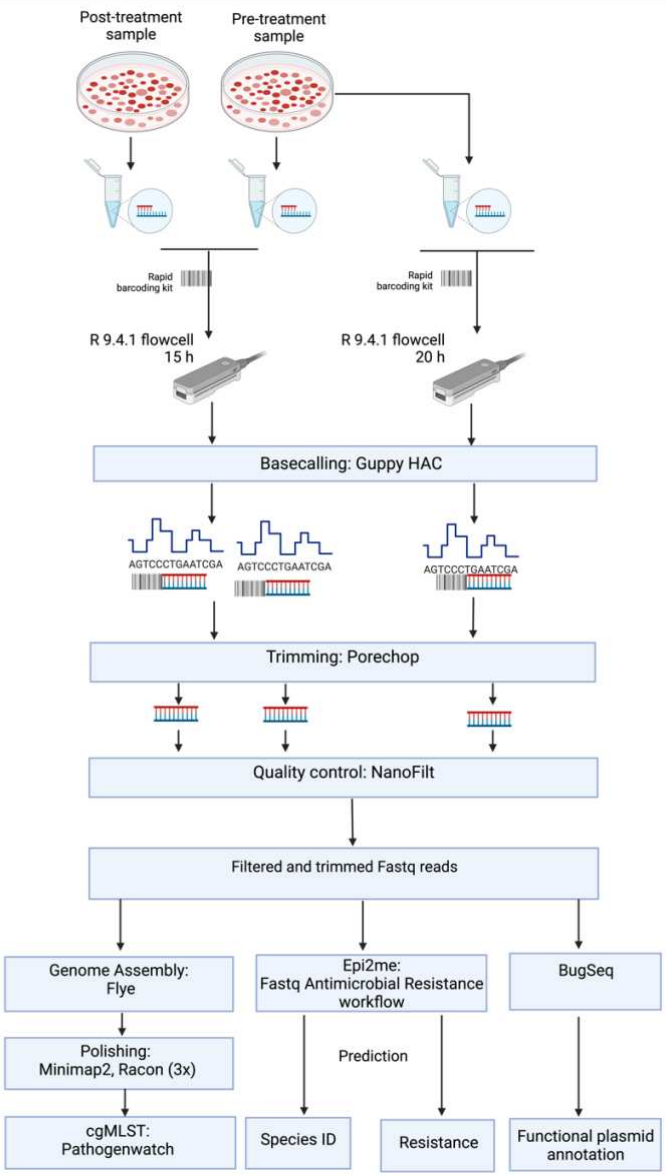
Nanopore sequencing workflow for species identification and resistance prediction

The stored isolates were thawed and grown overnight at 37°C on BD® Columbia blood agar plates. In the first round of sequencing, we isolated DNA from circa 10 CFUs of the pre-, and post-treatment samples using an automated magnetic-bead-based DNA purification approach with the Maxwell® RSC Blood DNA extraction protocol for gram-negative bacteria with the Promega Maxwell®RSC 48 Instrument (Promega Corporation, Madison, USA). DNA concentrations were measured with the Qubit™ (Thermo Fischer Scientific, Waltham, USA) dsDNA HS kit according to the manufacturer's instructions. Nanopore libraries were generated using the SQK-RBK004 Rapid Barcoding Kit and sequenced on an Oxford Nanopore Technologies MinION MK1b device with R9.4.1 flow cells for 15 hours as previously described[22], attaching one barcode per sample. The pore scan at the beginning of the sequencing run detected 781 pores with 378 remaining after the sequencing run was completed (MinKNOW version 22.10.5). For the second round of sequencing of the pre-treatment isolate, we increased the number of CFUs for DNA extraction to circa 50 and sequenced the extracted DNA for 20 hours on a new R.9.4.1 flow cell, maximising sequencing depth and coverage (**Supplementary Table 2**; **Supplementary Figure 1**). For this round of sequencing, 1651 pores were detected at the start and 1129 at the end of sequencing.

Supplementary Table 2. Real-time genomic sequencing and assembly metrics before and after treatment with CAZ-AVI. Genomic material of each bacterial isolate was sequenced for 15 hours; the pre-treatment isolate was additionally sequenced for a further 20 hours to simulate adaptive sequencing in the clinical setting.

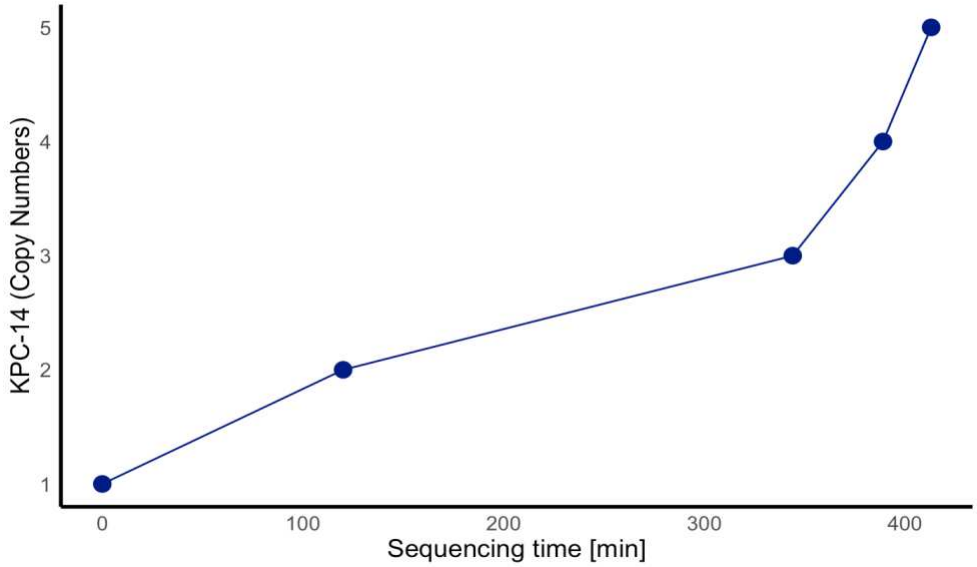
Genomic sequencing and assembly metrics	Pre-treatment sample		Post-treatment sample
	15h	+20h	15h
Median read quality (Phred)	12.3	13.0	12.3
Median read length [kb]	4.1	5.3	4.5
Total number of bases [Mb]	442	1,551	194
Assembly coverage (median)	68	271	30
Number of contigs (>50 kb)	5	5	5
Contig N50 [Mb]	5.2	5.2	5.3

For both sequencing rounds, fast5 files were basecalled with Guppy v6.3.2 using the "High-accuracy" model (r9.4.1_450bps_hac) [23], resulting in fastq files for each read. We then used Porechop version 0.2.3 (https://github.com/rrwick/Porechop) to trim the adapters and filtered out low-quality reads with a threshold quality score of 9 and short reads (<200b) with Nanofilt version 2.8.0 (https://github.com/wdecoster/nanofilt). Sequencing summaries were generated for each barcode using NanoStat v1.6.0 (https://github.com/wdecoster/nanostat[24]). We identified resistance-conferring genetic patterns with the EPI2ME Fastq Antimicrobial Resistance (v2023.04.26–1808834) workflow. This workflow includes quality control of the filtered and trimmed nanopore reads, taxa identification by the WIMP (What's in My Pot; v2023.06.13-1865548) workflow based on the NCBI RefSeq database and Centrifuge[25] to then analyse classified nanopore reads for resistance genes using The Comprehensive Antibiotic Resistance Database (CARD, https://card.mcmaster.ca/home, accessed 14. May 2023[26]). We then retained resistance gene matches identified via the protein homolog model, which is the most conservative model of the Fastq Antimicrobial Resistance workflow[26]. Additionally, we chose to only include respective KPC-2 and KPC-14 resistance genes with detection accuracies above 90% to minimise the chances of false positive resistance gene detection.



Supplementary Figure 1. Bioinformatic workflow for species identification and antibiotic resistance prediction from nanopore sequencing data of pre- and post-treatment isolates (HAC: High-accuracy basecalling; ID: Identification; cgMLST: core genome Multi-Locus Sequence Typing).

To make inferences on KPC-14 copy number detection over time, we further matched the KPC-14 alignments provided by the Fastq Antimicrobial Resistance workflow to the timepoints of the corresponding reads in the filtered and trimmed fastq file (**Supplementary Figure 2**).

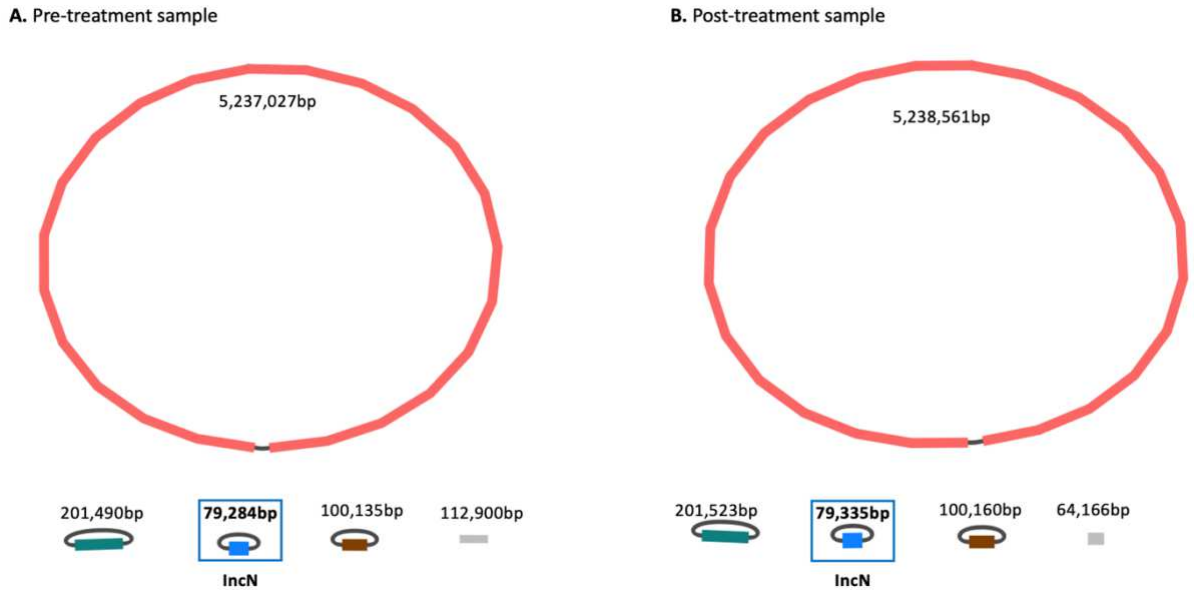


Supplementary Figure 2. Timeline of KPC-14 copy number detection during simulation of adaptive sequencing in the clinical setting applied to the pre-treatment bacterial isolate. The run started with one KPC-14 copy number which was obtained in the first sequencing run of the pre-treatment isolate. The second KPC-14 copy was found after two hours, and all additional KPC-14 copies were detected within the first seven hours of sequencing.

Core genome multi-locus sequence typing and chromosome/plasmid alignment analysis

We assembled the reads into contigs to prepare for core genome multi-locus sequence typing (cgMLST), which is a more stringent method for strain-level detection tailored to whole genome sequencing data compared to the seven locus MLST scheme[27,28]. In line with previous benchmarking studies for nanopore-only assemblies[22,29], we assembled the bacterial genome using the nano-corr option from Flye v2.9.1 (https://github.com/fenderglass/Flye). To further increase the accuracy of our assemblies, we realigned the reads to the assembled contigs using Minimap2 v2.18[30] followed by three polishing rounds with Racon 1.5 (https://github.com/isovic/racon, **Supplementary Figure 1**). Assembly sequencing depth was assessed with SAMTools depth v. 1.18 (https://github.com/samtools/samtools) after converting the polished contigs to sorted BAM files. We analyzed our *de novo* assemblies using the Pathogenwatch v2.3.1[28] platform which integrates Kleborate[27], facilitating *Klebsiella* species complex assignments, along with the identification of acquired virulence factors and recognized resistance markers, including both acquired genes and chromosomal mutations. Additionally, Pathogenwatch implements the Life Identification Number (LIN) code scheme, which enables sublineage and clonal groups assignment based on cgMLST profile comparisons[28]. Both isolates had a virulence score of 0/5 according to the Kleborate platform.

For more insight into the plasmid structures, we visualised the chromosomal and plasmid genome assemblies (**Supplementary Figure 3**) using Bandage v0.90 (https://github.com/rrwick/Bandage). Additionally, we uploaded the sequencing reads to BugSeq (https://bugseq.com/academic) for automated functional annotation of the plasmid genomes including resistance gene detection against a curated plasmid database[31]. Here, the IncN plasmid was identified as the carrier of the KPC-2 and KPC-14 genes respectively. To assess similarity between the IncN plasmids and the KPC-2 and KPC-14 genes on the plasmids, we aligned the respective IncN plasmids against each other, as well as the KPC gene sequences to the respective plasmids using the local alignment tool blastn provided by NCBI (https://www.ncbi.nlm.nih.gov; BLAST v2.14.1+).



Supplementary Figure 3. Genome and plasmid *de novo* assemblies and respective contig lengths. *The K. pneumoniae* chromosome is highlighted in red and the IncN plasmid is highlighted by the blue square. Additional circular and non-circular accessory genomes are visualised in green, brown, and grey. **A.** Pre-treatment sample. **B.** Post-treatment sample.

Reference Center annotation of novel KPC variant

Upon submission of the post-treatment bacterial isolate to the German National Reference Center for Gram-Negative Bacteria (https://memiserf.medmikro.ruhr-uni-bochum.de/nrz), the KPC resistance gene that our workflow defined as KPC-14 was identified as a previously unseen KPC and named *KPC-159* (NCBI sequence ID: OQ450354.1). Using blastn, we confirmed that this KPC-159 gene shows 99.9% overlap in protein sequence (875/876 amino acids) with KPC-14,.

Data availability

All FAST5 nanopore sequencing data has been uploaded to NCBI (https://www.ncbi.nlm.nih.gov, SRA submission: SUB13974279).

Code availability

All computational scripts for processing the FAST5 data are available here: https://github.com/Genomics4OneHealth/Klebsiella_pneumoniae_KPC_nanopore_sequencing

References

20. The European Committee on Antimicrobial Susceptibility Testing. Breakpoint tables for interpretation of MICs and zone diameters, version 13.1, 2023. [Internet]. [cited 2023 Oct 4]. Available from: https://www.eucast.org/clinical_breakpoints.
21. Hong, J., Kang, D. & Kim, D. Performance Evaluation of the Newly Developed In Vitro Rapid Diagnostic Test for Detecting OXA-48-Like, KPC-, NDM-, VIM- and IMP-Type Carbapenemases: The RESIST-5 O.K.N.V.I. Multiplex Lateral Flow Assay. *Antibiotics* 10, 460 (2021).
22. Wick, R. R., Judd, L. M., Gorrie, C. L. & Holt, K. E. Completing bacterial genome assemblies with multiplex MinION sequencing. *Microb Genom* 3, (2017).
23. Oxford Nanopore Technologies, *Guppy v6.3.2*. [cited 2023 Oct 4]. Available from: https://github.com/nanoporetech/pyguppyclient.
24. De Coster, W., D'Hert, S., Schultz, D. T., Cruts, M. & Van Broeckhoven, C. NanoPack: visualizing and processing long-read sequencing data. *Bioinformatics* 34, 2666–2669 (2018).
25. Kim, D., Song, L., Breitwieser, F. P. & Salzberg, S. L. Centrifuge: rapid and sensitive classification of metagenomic sequences. *Genome Res* 26, 1721–1729 (2016).
26. Alcock, B. P. *et al.* CARD 2023: expanded curation, support for machine learning, and resistome prediction at the Comprehensive Antibiotic Resistance Database. *Nucleic Acids Res* 51, D690–D699 (2023).
27. Lam, M. M. C. *et al.* A genomic surveillance framework and genotyping tool for *Klebsiella pneumoniae* and its related species complex. *Nat Commun* 12, 4188 (2021).
28. Argimón, S. *et al.* Rapid Genomic Characterization and Global Surveillance of *Klebsiella* Using Pathogenwatch. *Clinical Infectious Diseases* 73, S325–S335 (2021).
29. Foster-Nyarko, E. *et al.* Nanopore-only assemblies for genomic surveillance of the global priority drug-resistant pathogen, *Klebsiella pneumoniae*. *Microb Genom* 9, (2023).
30. Li, H. Minimap2: pairwise alignment for nucleotide sequences. *Bioinformatics* 34, 3094–3100 (2018).
31. Chandrakumar, I. *et al.* BugSplit enables genome-resolved metagenomics through highly accurate taxonomic binning of metagenomic assemblies. *Commun Biol* 5, 151 (2022).

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Contributions

LU and ES conceptualised the project. ES, LU and EFN designed the experiments. ES, TR, and AP conducted the real-time sequencing experiments under LU's supervision. ES, TR, and SV conducted the bioinformatic analysis including data curation, formal analysis, and visualisation, under LU's and EFN's supervision. NG consulted on genomic data analysis and interpretation. NW and CC supervised the clinical case and the phenotypic testing. CPC and NW provided samples and laboratory equipment. ES and LU wrote the manuscript with input from all co-authors.

Competing interests.

We declare no competing interests.

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Ethics declarations

Ethical approval was given by the ethics committee of the Technical University of Munich, Germany (2023-575-W-NP, 2022-611-S-KH). Informed patient consent was waived as samples were taken under routine diagnostics. The research conformed to the principles of the Helsinki Declaration.